

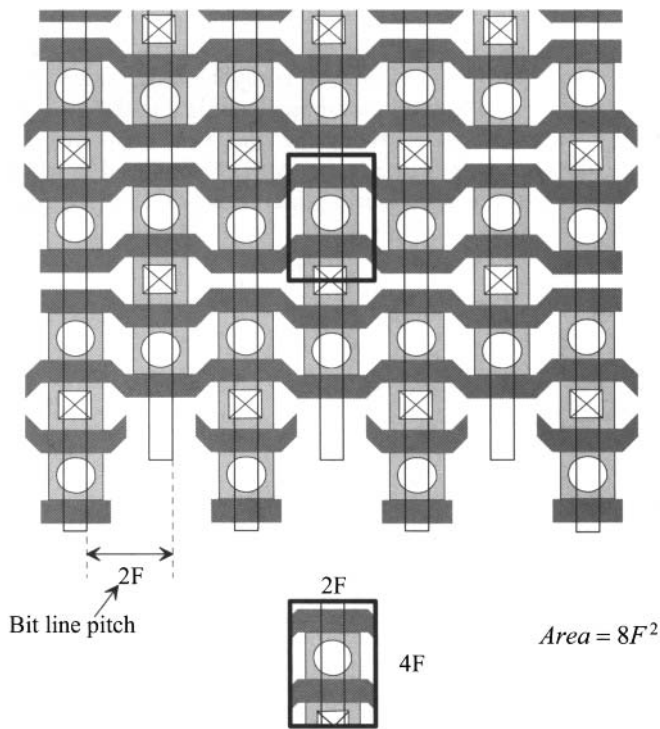


Figure 16.21 Folded architecture layout and cell size.

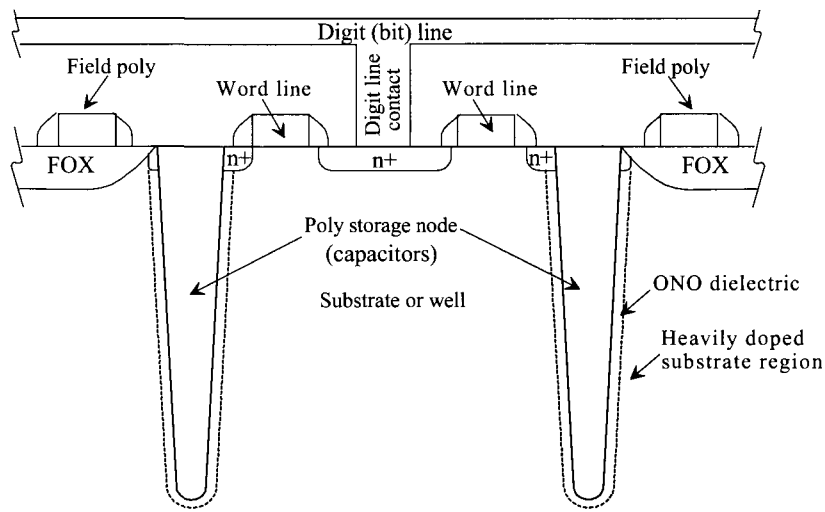


Figure 16.22 Cross-sectional view of a trench capacitor cell.

hole's depth to its diameter) can be quite high, which leads to processing concerns. In Fig. 16.23 the buried capacitor cell pair is seen. Unlike the trench capacitor-based cell that places the cell directly in the substrate, the capacitor is "buried" under the digit line but still above the substrate. The benefit of this cell is simpler processing steps. One problem with this cell, when compared to the trench-based cell, is that the parasitic capacitances are higher (such as the capacitance loading the (digit) bit line). Also, for a given bit capacitance, C_{mbit} , the area of the buried capacitor cell may need to increase, while the area of the cell using the trench capacitor remains constant. The depth of the trench capacitor is increased for more capacitance.

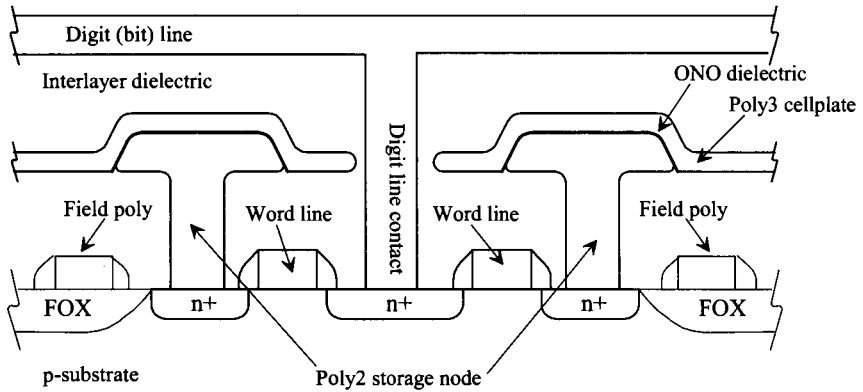


Figure 16.23 Cross-sectional view of a buried capacitor cell.

As we saw in Eq. (16.7), the value of C_{mbit} is very important. Thin dielectrics with high dielectric constants are used to implement C_{mbit} . Tricks like roughing up the dielectric to increase the surface area are often employed. To minimize the stress on the oxide, the common node of the capacitor is usually tied to $VDD/2$, as seen in Fig. 16.24. When a logic 1 (VDD) is written to the cell, the charge on the capacitor is $(VDD/2) \cdot C_{mbit}$. When a logic 0 (ground) is written, the charge on the capacitor is $-(VDD/2) \cdot C_{mbit}$. The difference in these charges is $VDD \cdot C_{mbit}$, which is the same difference we would have if the common node were connected to ground.

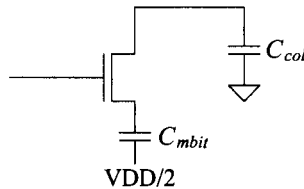
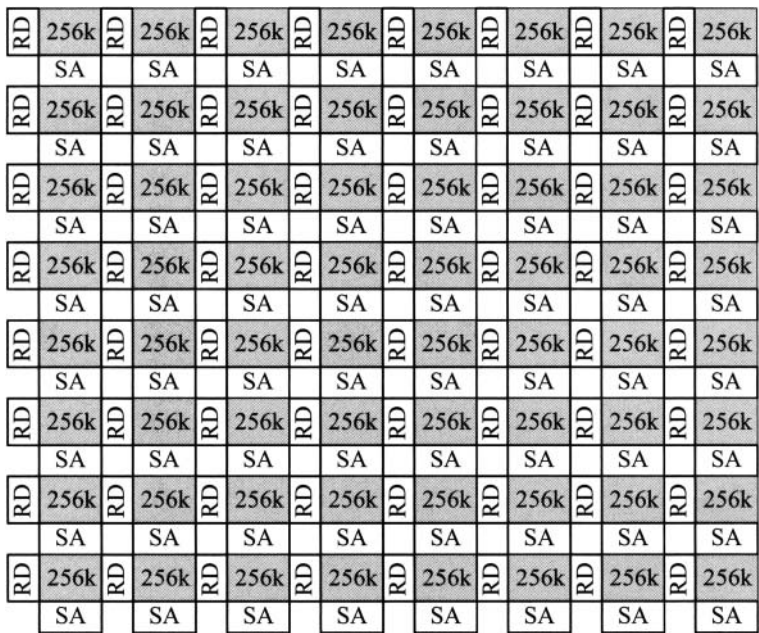


Figure 16.24 Holding the capacitor's common plate at $VDD/2$ to minimize oxide stress.

16.1.3 Chip Organization

The number of rows in a DRAM is limited because each additional row adds capacitance to the bit line. The power goes up, as seen in Eq. (16.4), with increasing C_{col} . The number of bit lines is limited by the delay through the row line (the length of the row line), as indicated in Eq. (16.9). For these reasons, the size of a DRAM array is limited. A typical array size is 512 word line pairs and 512 bit lines for a total memory size of 256 kbits. We use the term “pairs” to indicate that, as seen in Fig. 16.20, there are actually 1,024 row lines in a 256k array. We can arrange these 256k arrays to form a larger memory array. Figure 16.25 shows the basic idea for an 16-Mbit array block. A 1-Gbit DRAM uses 8,192 256k arrays. These arrays may be further subdivided into “banks” of memory. We might, for a 1-Gbit memory, have 8 banks of 128-Mbits. We subdivide the memory so that an operation (say a read) can be done in one bank while, at the same time, some other operation (say a refresh) can be performed in a different bank.

The *array efficiency* is defined as the ratio of the area of the memory blocks to the total chip area. A typical value ranges from 50 to 60%. The peripheral circuitry seen in Fig. 16.25 (the sense amplifier and row decoder blocks) take up a significant amount of chip real estate. Because of their importance, we’ll devote the entire next section to the design of peripheral circuits including sense amplifiers, row drivers, and decoder circuits.



 Row decoder and driver circuitry

 Sense amplifier and column decoder circuitry.

Figure 16.25 A 16-Mbit array block.

16.2 Peripheral Circuits

In this section we discuss the design of general sense amplifiers (for general use in memory design), row drivers (remembering, for a DRAM, the row voltage needs to be driven above V_{DD}), and column and row decoder circuits.

16.2.1 Sense Amplifier Design

Examine the clocked sense amplifier seen in Fig. 16.26. When *clock* is low, MS3 is off while MS1 and MS2 are on. Our input signals can't go below V_{THP} (with body effect) without shutting MS1 and MS2 off. Assuming that the inputs stay above V_{THP} , the drains of M1/M3 and M2/M4 are charged to $In+$ or $In-$ (creating an imbalance). When *clock* goes high, the imbalance causes the circuit to latch high or low depending on the state of the inputs. Figure 16.27 is the simulation output showing the typical operation of the circuit. When *clock* is low, the circuit outputs are not valid logic signals but rather, ideally, track the input signals. This circuit is plagued with problems including: kickback noise, memory, and significant potential contention current.

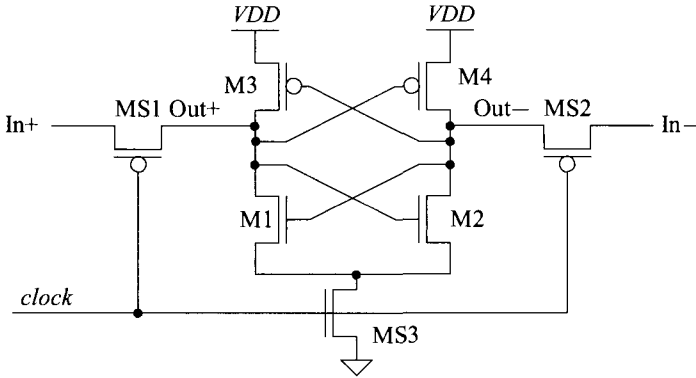


Figure 16.26 Clocked sense amplifier.

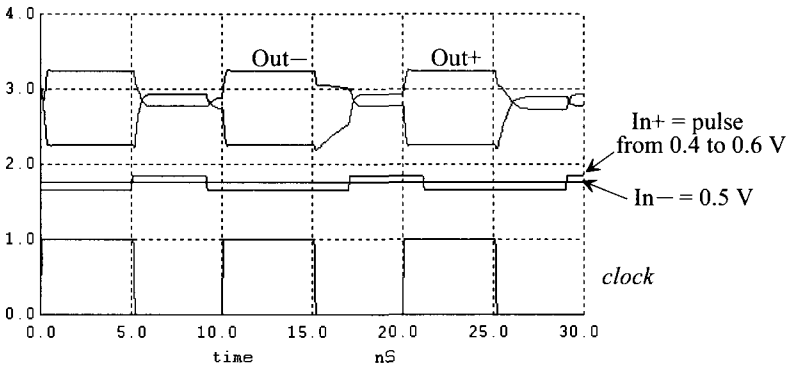


Figure 16.27 Simulating the operation of the circuit in Fig. 16.26.

Kickback Noise and Clock Feedthrough

In the simulation that generated Fig. 16.27 we used, for our inputs, voltage sources. In a real application, the inputs would come from some other logic or a bit line (a source with a finite driving impedance). Consider including logic for driving the sense amplifier seen in Fig. 16.28. We use long-length inverters to simulate a weak driver circuit (such as a decoder made with a series of transmission gates). Seen in the figure are simulation results at the time when the clock goes high (we get similar glitches when the clock goes low). The large glitch seen in these figures is called *clock feedthrough* noise. It is present when a clock signal has a capacitive path directly to the input of the sensing or comparator circuit. The clock feedthrough noise in Fig. 16.28 is close to 50 mV on both inputs.

Another type of noise, called *kickback noise*, is present and injected into the inputs of the circuit when the latch switches states. *It's important to simulate the operation of the sense amplifier with nonideal sources (with finite source resistance) to determine the significance of clock feedthrough and kickback noise.* Using voltage sources with 10k resistors is a reasonable source for general circuit simulations. This noise is an important specification for a sense amplifier. It can often be the *limiting factor* when making sensitive measurements. If the kickback noise, for example, is too large, it can interfere with sensing. A simple example of a potential problem occurs when 256 sense amps are all being clocked at the same time (e.g., on the bottom of a memory array) and they are directly adjacent to each other. We don't want one of the amplifiers interfering with any of the others by feeding noise into the others' inputs.

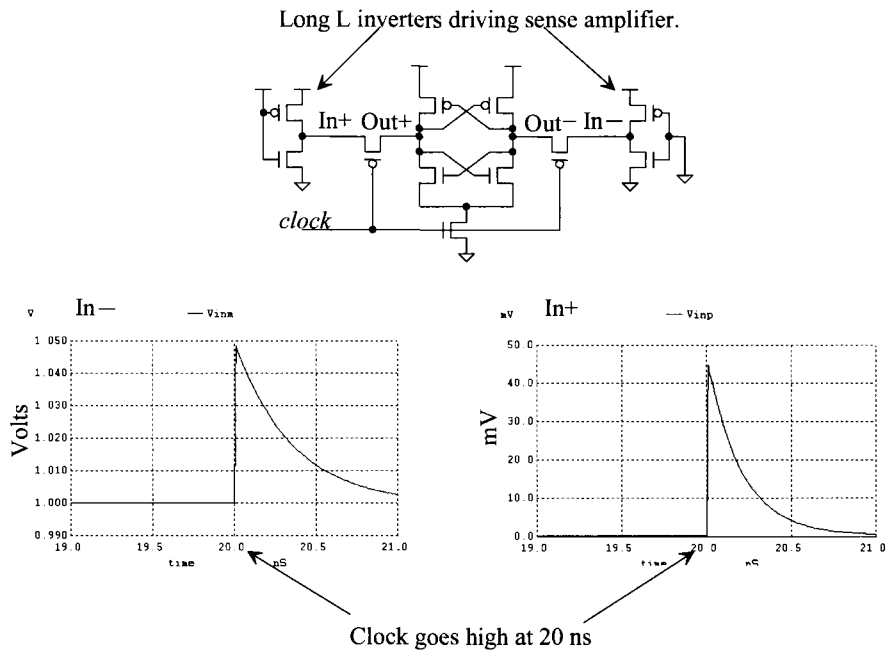


Figure 16.28 Clock feedthrough noise.

Memory

For a good comparison, the sense amplifier shouldn't have a memory of previous sensing operations. For the sense amplifier in Fig. 16.26, the outputs are actively driven by the input signals prior to clocking. However, the drain of MS3 (sources of M1/M2) is floating. This node is a dynamic node that floats to a voltage dependent on the previous decision and the input signals. *For a precision sense operation, all nodes in the sense amplifier must be driven or equilibrated, prior to clocking, to a known voltage.*

Current Draw

Because there may be thousands of sense amplifiers operating on a chip at the same time, it is extremely important to minimize the power used by these circuits. Let's take a look at the current supplied by V_{DD} to the circuit in Fig. 16.26 (with the signals seen in Fig. 16.27). Because the inputs actively drive the gates of M3 and M4, it is possible M3/M4 source a significant current into the inputs. In this case, Fig. 16.27, the V_{SG} voltages are roughly 0.5 V (and so both M3 and M4 are conducting a drain current). The current supplied by V_{DD} is seen in Fig. 16.29. *Clock* is high during 0 to 5ns, 10ns to 15ns, etc. (when the current supplied by V_{DD} is small). If the average current is 50 μA and there are 1,000 sense amplifiers operating on the chip, then V_{DD} supplies 50 mA of current (just to the sense amplifiers). *For minimum power, it's important that there are no DC paths from V_{DD} to ground except during switching times.*

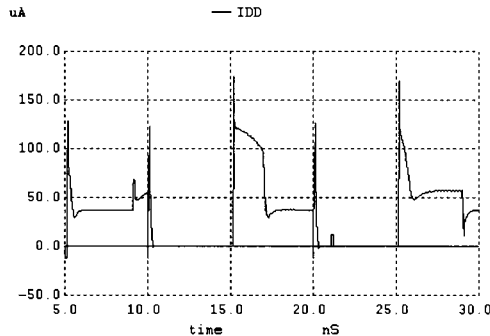


Figure 16.29 The current that flows in the sense amplifier in Fig. 16.26.

Contention Current (Switching Current)

When switching takes place in the sense amplifier of Fig. 16.26, both M3 and M4 are conducting. When *clock* goes high, both M1 and M2 turn on and conduct current as well. In this situation there is a direct path from V_{DD} to ground. If the inputs are at relatively the same voltages, the time that M1–M4 remain conducting can be very long (resulting in significant current being pulled from V_{DD}). To eliminate this *metastable* condition and force the comparator to make a decision (switch to valid logic levels), the gain in series with the input signals can be increased. A simple example of increasing input signal gain is to place an amplifier on the inputs of the latch.

Note that if our input signals amplitudes are within a V_{THP} of V_{DD} , the PMOS devices are off and we don't have this problem. The inputs can generate an imbalance on

the drains/gates of the NMOS transistors when clock goes high. Neither PMOS device will turn on until either M1 or M2's drain moves below $V_{DD} - V_{THP}$. For low power (low contention current), try to keep one side of the latch shut off until a significant imbalance is present in the circuit.

Removing Sense Amplifier Memory

Figure 16.30 shows how the sense amplifier's memory can be erased. M1–M4 form a latch (Fig. 13.16). To remove the sense amplifier's memory, *all* nodes in the sense amplifier must be actively driven to a known voltage (no floating or dynamically charged nodes). When *clock* is low, the sense amplifier's outputs are pulled to V_{DD} through MS3 and MS4. The MOSFETs MS1 and MS2 are off, breaking the connection between V_{DD} and ground (so no current flows in the latch). The gates of M1 and M2 are at V_{DD} so their drains are actively driven to ground. In other words, when *clock* is low, all nodes in the circuit are either pulled high to V_{DD} or low to ground.

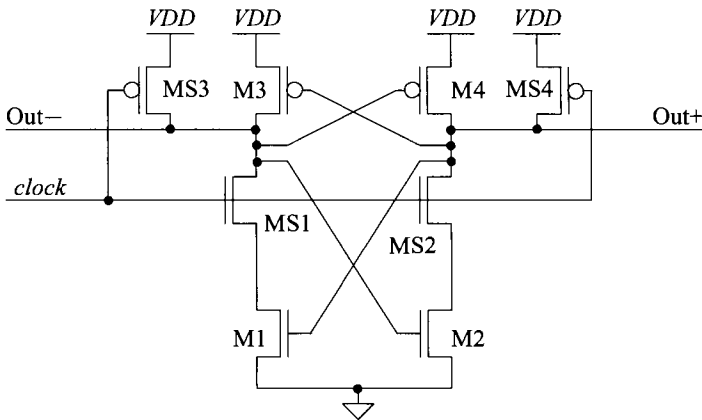


Figure 16.30 Removing memory in a sense amplifier.

Creating an Imbalance and Reducing Kickback Noise

When *clock* goes high, the latch action can take place. For the comparison to function as desired, we need to generate an imbalance. This, alone, isn't too difficult. What is difficult is creating an imbalance for a wide range of input signals while not causing an excessive amount of current to flow in the circuit. Consider the additions to our sensing circuit seen in Fig. 16.31. MB1/MB2 can be connected to either the drain or the source of MS1/MS2, as seen in the figure. Let's first consider connecting them to the drains. When *clock* is low, the drains of MB1 and MB2 are pulled to V_{DD} . If the input voltages are significant, a *large* current can be drawn from V_{DD} (when *clock* is low). If the input voltages are relatively small, then the current pulled from V_{DD} may be tolerable (keeping in mind that the sense amplifier doesn't function correctly when the input voltages are less than V_{THN}). The benefit of connecting to the drains is high gain. Very small voltage differences can cause quick sensing (the outputs swing quickly to valid logic voltages). Note that the possibility for large current flowing through either MB1/MB2 is present independent of the state of *clock*.

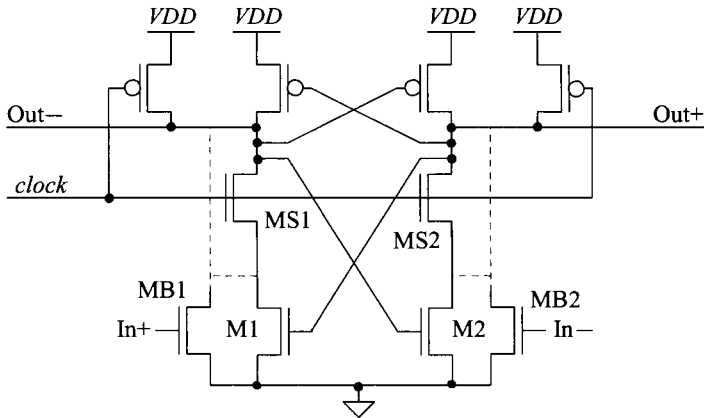


Figure 16.31 Creating an imbalance in a sense amplifier.

Next let's consider connecting MB1/MB2 to the sources of MS1/MS2 (drains of M1/M2). When *clock* goes low, the drains of MB1/MB2 are pulled low through M1/M2. No current flows in the comparator when *clock* is low (unlike when the drains of MB1/MB2 are connected to the drains of MS1/MS2). The gain is much lower because now MB1/MB2 are operating in the triode region when *clock* goes high. The potential for large current flow is still present when *clock* is high but, now, the maximum value on the drain of MB1/MB2 is $V_{DD} - V_{THN}$. And so the potential for MB1/MB2 to move into the triode region can result in a smaller output current.

We seem to have a problem. To reduce the possibility of metastability, we need to increase the gain in series with the signal path (increase the W/L ratio of MB1 and MB2; that is, reduce their switching resistance). However, this results in larger power dissipation. How can we keep power down while maximizing the sensitivity of our sense amplifier? While we can use long L devices for MB1 and MB2 so they never pull significant current (at the cost of sensitivity and speed) or have a separate pre-amplifier to generate the input signals, let's look at some other ideas (noting that no design is perfect but a trade-off between power, speed, and sensitivity).

Figure 16.32 shows one idea for creating an imbalance without the possibility of significant current flow. MB1 and MB2 are used to create an imbalance in the gate-source voltages of M1/M2. Since, prior to switching, the voltage on the gates of M1/M2 is V_{DD} , we can still have good sensitivity even though MB1 and MB2 are operating in the triode region (look like resistors). The large voltage dropped across $V_{GS,M1}$ and $V_{DS,MB1}$ ensure that good gain. It is important, however, that the inputs signals are $> V_{THN}$ to ensure that the circuit operates properly. The sensing fails if this isn't the case because neither MB1 or MB2 can turn on and provide the needed path to ground for proper logic level signals. As long as the inputs are above the threshold voltage, there aren't any dynamic nodes (*no memory in the sense amplifier*). The sources of MS1 and MS2 are still pulled to ground when *clock* is low. Kickback noise is reduced using this topology because the inputs are isolated from the latch by MB1/MB2. Kickback noise is still present and must be considered when designing and simulating this sense amplifier.

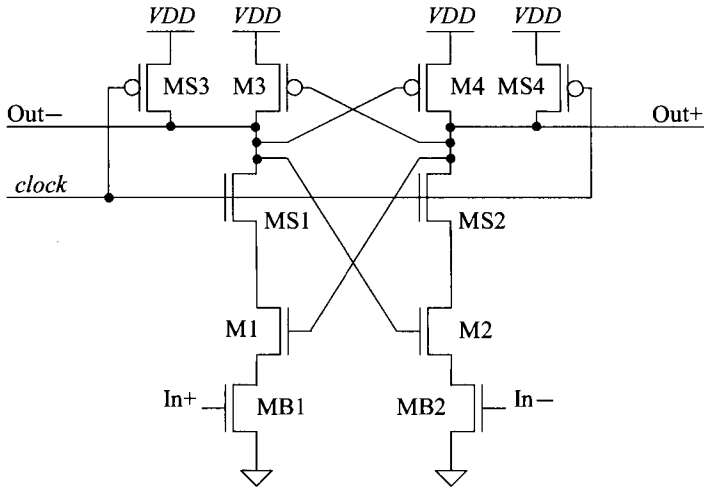


Figure 16.32 Reducing power in a sense amplifier.

Note that we might try to design our comparator to operate over a wide range of input voltages by adding PMOS devices in series with the sources of M3/M4, as seen in Fig. 16.33. We know that MB1–MB4 must be capable of turning on when *clock* goes high. If either pair can't turn on, then the latch won't be able to generate full logic levels. For the design in Fig. 16.33, the input signals would have to fall within V_{THN} and $VDD - V_{THP}$ (more restricted than the circuit in Fig. 16.32). Also, and perhaps more importantly, MB3/MB4 can't generate an imbalance in the latch. If the sources of M1/M2 are connected to ground so that MB1/MB2 don't generate the imbalance, then when *clock* goes high, MB3 and MB4 are off. They won't turn on until the NMOS pair M1/M2 turns on and drops the gates of M3/M4 below $VDD - V_{THP}$. For great differences in the input signals, the comparator may still function correctly. However, for the majority of the input signal differences, the resulting output logic levels is determined by the matching between M1/M2, e.g., the one with the lower "actual" V_{THN} will turn on faster.

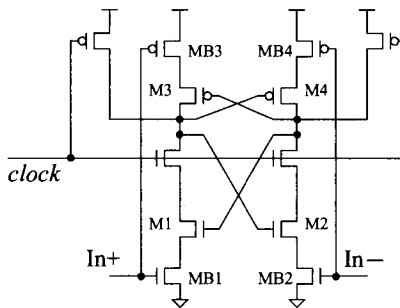


Figure 16.33 A bad design: how not to get wide-swing operation.

Example 16.1

Suppose that it is suggested that the gates of M3/M4 be tied to the drains of M1/M2 or MB1/MB2, respectively, in the sense amplifier (clocked comparator) seen in Fig. 16.32. Comment on the concerns.

If the gates of M3/M4 are tied to the drains of M2/M1, then, when *clock* goes high, a significant contention current flows in the latch. As discussed earlier, it is desirable that one side of the latch (either the NMOS or PMOS pairs) is off when the latch is clocked in order to reduce power. The same concerns are present when these gates are connected to the drains of MB1/MB2.

We might try to eliminate MS3/MS4 when we tie the gates of M3/M4 to the drains of MB1/MB2. Because the drains of MB1/MB2 always move to ground when *clock* is low (again assuming the inputs are $> V_{THN}$), M3/M4 will turn on and pull the outputs high. When *clock* goes high, though, it's impossible to shut M3/M4 off. Because the maximum voltage on the sources of MS1/MS2 is $V_{DD} - V_{THN}$, the gates of M3/M4 can't go to V_{DD} to shut one of the devices off. ■

Increasing the Input Range

Figure 16.34 shows adding, to our basic sense amplifier in Fig. 16.32, MOSFETs MB3–MB8 so that the circuit will function with input signals ranging from 0 to V_{DD} (actually it will operate correctly with input signals beyond the power supply rails). We know current can only flow out of the sources of M1/M2, so our additional circuitry, for creating the imbalance, must be capable of sinking current from these sources (and so we'll add MB3 and MB4 for this reason). Next, we know that MB1/MB2 function fine for creating the imbalance if the input signals are above V_{THN} . However, if the input signals are below V_{THN} , they turn off. To level-shift the input, let's use the MB5–MB8. If the input signals are $< V_{THN}$, then MB7 and MB8 are on. A difference in the input voltages causes different currents to flow in MB5 and MB6. The different currents flowing in these transistors results in different voltages across each one. This voltage difference is then used to create an imbalance in MB3 and MB4. Unfortunately, the addition of MB5–MB6 contradicts our earlier statement that, “*For minimum power it's important that there are no DC paths from V_{DD} to ground except during switching times.*” There is a DC path through MB8/MB6 and MB7/MB5. To lower the power, we can increase the length of MB5 and MB6 so that the continuous current flowing through the DC path is lessened.

Simulation Examples

Let's simulate the operation of the circuit in Fig. 16.32. We know that the outputs of the sense-amp go high every time *clock* goes low. To make the outputs change only on the rising edge of *clock*, we can use the SR latch discussed in Ch. 13 on the output of the circuit, Fig. 16.35. Now, with the addition of the NAND gates, when the sense amplifier's outputs are high, the outputs of the latch don't change from the previous decision (made on the rising edge of *clock*).

Figure 16.36 shows the current supplied by V_{DD} for the circuit in Fig. 16.35. While the current seen in this figure includes the current supplied to the NAND gates, it still should be compared to Fig. 16.29. Notice in Fig. 16.36 that current flows only during the times the clock changes states (unlike the current flow in Fig. 16.29).

